

Valentin Gaucher

+1 720 487 6039 valgau01@gmail.com Google Scholar Personal Website in Valentin Gaucher

About Me

Robotist and aerospace engineer working on data-driven and nonlinear MPC for aerial robots in contact-rich, interactive environments. I leverage advances in computation and embedded electronics to build robotic systems that are safer, smarter, and more agile. **Anticipated Graduation: Spring 2028. French citizen, F-1, CPT-eligible.**

Education

PhD Arizona State University (USA) , Optimal Control & Robotics	Aug. 2024 to Spring. 2028
M.T California Institute of Technology (USA) , Master Thesis, Aerospace & Robotics	Apr. 2023 to Apr. 2024
M.S ISAE Supaero (France) , Aerospace Engineering, Dynamics & Control	Sep. 2021 to Sep. 2023
V.S Politecnico di Milano (Italy) , Visiting Student, Aerospace & Control	Feb. 2021 to Sep. 2021
B.S ENSTA (France) , Mechanical Engineering	Sep. 2018 to Sep. 2021

Additional Training

ETH Zürich , Autonomous Mobile Robots Online Course (Computer Vision, SLAM)	Jan. 2024, 6 months
NASA Jet Propulsion Laboratory , DARTS Lab Workshop (Flexible Multibody Dynamics)	Aug. 2023, 1 week

Relevant Experience

Robotics and Intelligent Systems Laboratory (RISE), ASU , Graduate Research Assistant <i>Developing control algorithms to solve trajectory optimization problems in drone-environment interactions: Data Driven and Non-Linear MPC, hybrid dynamical systems, optimal control for UAVs in contact-rich applications.</i>	Tempe, USA Aug. 2024 to Now Current
Centre National d'Etude Spatiales (CNES) , Mars Rover Operations Engineer <i>Real-time operation and commanding of ChemCam (MSL) and SuperCam (Mars 2020) science instruments aboard active Mars rovers. Responsible for uplink sequence validation, instrument health monitoring, and anomaly response in a fault-critical planetary robotics environment at the CNES Toulouse Space Center (French Space Agency).</i>	Toulouse, France April 2024 to August 2024 6 Months
NASA Jet Propulsion Laboratory / Caltech CAST , Robotics, Visiting Student Researcher <i>Experimental characterization of Mars Science Helicopter takeoff performance from a jet-pack. Takeoff dynamics & control, development of a real-time robust controller. Testing at Caltech GALCIT CAST facility.</i>	Pasadena, USA Apr. 2023 to Nov. 2023 8 months
AerospacePoliMi , Control, Visiting Student <i>Quadrotor robust attitude controller using H_∞ and Monte Carlo validation. Studied aero-elasticity effects on quadrotors for robust control purposes.</i>	Milan, Italy Feb. 2021 to Aug. 2021 6 months

Research Interests

Aerial robotics for extreme environments: Deployment, sensing, and control of aerial robots in extreme environments. Investigating tactile based planning [6] and sensing methods for planetary exploration [4], sample recovery and search and rescue.

Contact-Rich Optimal Control and Representation: Planning and control that let dynamical systems survive, leverage or recover from contact. This includes collision recovery [1], contact-aware optimal planning [2][6], landing and takeoff [2][3], and contact modeling [1].

Real-time control and experimental validation: Deployment of real-time optimal control methods on flight hardware and high-fidelity simulation (MuJoCo, ROS2, PX4), with a focus on closing the simulation-to-real gap [3]–[7].

Research Software & Tools

MuJoCo-PX4 Simulation and Deployment Pipeline: Developed a MuJoCo-based simulation environment running the full ROS2-PX4 stack as an alternative to Gazebo, supporting configurable obstacle courses, vision sensors and IMU for rapid prototyping of optimal control algorithms (see [GIT](#)). Currently extending it, with a supervised intern, as a deployment agent that automatically ports simulation-side controllers to the ROS2-PX4 stack for flight on real hardware. Used as the teaching platform for the graduate course *System Control and Optimization* at ASU.

Darvis - Natural Language Assistant for Drone Experiments: Originated and lead contributor of an LLM-based assistant for live ROS2-PX4 flight sessions, allowing operators to modify setpoints and trigger deployments by voice or text, extract and query flight logs, and diagnose failed runs in natural language.

Mars Rotorcraft Simulation Environment: Developed a MuJoCo-based open-source simulator for Mars Rotorcraft (models proposed: Ingenuity, Chopper). Following HeliCAT structure, the vehicle sits inside a set of reduced-order models: ground effect, IMU and laser-altimeter error models, camera with lens distortion, vignetting, blur and sensor noise, dust-dependent illumination, and mean wind with turbulence. The code provide a PID, LQR and MPC control for trajectory tracking as well as a trained PPO for landing. Code available [here](#).

Papers

[1] **Published - IEEE Control Systems Letters (L-CSS), 2026:** *Dual Quaternion Based Contact Modeling for Fast and Smooth Collision Recovery of Quadrotors*. Valentin Gaucher & Wenlong Zhang. [IEEE link](#).

[2] **Accepted - IEEE Conference on Decision and Control (CDC), 2026:** *Switch-Time-Aware Hybrid Model Predictive Control Using the Hybrid Minimum Principle*. Valentin Gaucher & Wenlong Zhang.

[3] **Under Review - IEEE International Conference on Robotics and Automation (ICRA):** *Real Time Neural Model Predictive Control for Takeoff and Landing with Learned Ground Effect Scaling*. Valentin Gaucher, Yizhuang Garrard & Wenlong Zhang.

[4] **Under Review - IEEE Transactions on Field Robotics (T-FR):** *Dynamically Scaled Earth-Ambient Testing of Airload-Aware Takeoff for Mars Mid-Air Helicopter Deployment*. Marcel Veismann , Valentin Gaucher , Fernando Ruiz Vincueria , Peter Gavrilov , Luis Phillipe Tosi , Severin Schumacher, Filip Slezak, Katherine Tighe , Jacob Izraelevitz, Morteza Gharib , and Jeff Delaune.

[5] **Under Review - International Conference on Learning Representations (ICLR):** *Fast Strategy Solving for the Informed Player in Two-Player Zero-Sum Linear-Quadratic Differential Games with One-sided Information*. Mukesh Ghimire, Valentin Gaucher, Wenlong Zhang & Yi Ren.

[6] **In Preparation - IEEE Transactions on Robotics (T-RO) :** *Online Collision-Inclusive MPC for Quadrotors via Convex Relaxation of Hybrid Impact Law*. Valentin Gaucher & Wenlong Zhang.

[7] **In Preparation - IEEE Robotics and Automation Letters (RA-L) :** *Attitude-Aware Obstacle Avoidance for Agile Quadrotor Flight using Dual-Quaternion Model Predictive Control*. Valentin Gaucher & Wenlong Zhang.

[8] **In Preparation - Robotics: Science and Systems (RSS) :** *Collision Inclusive Trajectory Genertion with Convex Graph Search*. Valentin Gaucher & Wenlong Zhang.

Skills

Languages: Python (10 years experience), C/C++ (4 years experience), Matlab/Simulink (8 years experience)

Hardware: Jetson Thor, Jetson Orin Nano, Raspberry Pi 4/5, Arduino, PX4 Flight Controller

Software Libraries: Pytorch, ROS2, MuJoCo, Gazebo, Solidworks, CATIA, Abaqus

Optimization Libraries: acados, CasADi, OSQP, CVXPY, IPOPT, CUDA